

Saber Ayatollahi, CFA, MSc

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SUMMARY

Manager / Senior Manager, Quantitative Market Risk Models – Financial Engineering and Modeling candidate with ~7 years across quantitative model validation, derivatives pricing review, and model governance at Moody's Analytics, EY, and Ortec Finance. Holds delegated sign-off authority on valuation, sensitivity, and risk outputs for multi-asset institutional portfolios (\$5-25bn per engagement). Hands-on with stochastic scenario generation, Monte Carlo simulation, and Python analytics. CFA charterholder with dual MSc (Financial Modelling; Chemical Engineering – PDE and numerical-methods foundation); applied working knowledge of VaR/CVaR; familiar with FRTB and CCAR frameworks through client-advisory context, not direct trading-book capital submission.

CORE SKILLS

Quantitative Model Validation & Review • Derivatives Pricing Review (Rates, FX, Inflation) • Monte Carlo Simulation & Stochastic Scenario Generation • Value-at-Risk (VaR) & Conditional VaR (CVaR) • Yield Curve Construction & Calibration • Interest Rate Risk Analytics (Parallel & Non-Parallel Shocks) • Stress Testing & Scenario Analysis • Model Governance, Sign-Off & Documentation • Python (Advanced), SQL (Intermediate), MATLAB (Intermediate), R (Intermediate) • PDE & Numerical Methods (Chemical Engineering MSc)

EXPERIENCE

Moody's Analytics

Asst. Director - Modelling Services (Investment & ALM Risk Lead) | Toronto, May 2023 - Present

Derivatives Pricing Review & Quantitative Model Validation

- Holds delegated sign-off authority on valuation, sensitivity, and risk outputs delivered to institutional client portfolios ranging \$5-25bn per engagement (cumulatively ~\$50bn+); independently reviews curve construction, spread calibration, and cross-asset interactions prior to production release.
- Validates derivatives pricing outputs across rates, FX, and inflation – reviews sensitivity consistency and scenario impacts at portfolio-level aggregates; escalates outputs lacking mathematical or economic defensibility.
- Reviews interest rate risk and duration analytics under parallel and non-parallel rate shocks; applies quantitative methodologies analogous to sensitivities-based market-risk frameworks.

Model Governance, Documentation & Stakeholder Communication

- Operates within a formal model-governance framework – reviews assumptions, validates economic defensibility, and documents to standards consistent with SR 11-7 and OSFI E-23 expectations.
- Serves as escalation point for client-critical analytical issues, interfacing with Heads of Risk and investment stakeholders at Canadian and U.S. institutional clients.
- Prepares quantitative summaries for senior stakeholders on interest rate exposure, scenario impacts, and sensitivity behaviour; translates model outputs into committee-ready narratives.

Quantitative Model Development & Python Tooling

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios – embedding behavioral assumptions, prepayment logic, and macro stress overlays.
- Re-engineered manual spreadsheet workflows into scalable, auditable Python analytics pipelines with embedded validation, logging, and version control.
- Built agentic AI development workflows (Claude Code, Cursor IDE) automating code generation, validation scaffolding, and anomaly detection – reducing development cycle time by an estimated 30-40% on comparable modules.

Client Service Specialist - PFaroe Analytics Platform | Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform; advised on asset, liability, and risk-model configurations and validated model outputs post-deployment.

- Contributed to the successful migration of all assigned client accounts from the legacy Calypso platform to PFaroe PM, translating client investment-team requirements into structured Product Owner development requests.

Ernst & Young (EY)

Senior Consultant - IFRS & Financial Risk Transformation | Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 model and technology transformation projects for Canadian and international insurance clients, aligning regulatory requirements with system implementation and financial-reporting frameworks.
- Coordinated across actuarial, finance, IT, risk, and program-management teams on milestone delivery, governance documentation, and implementation materials.

Ortec Finance Canada

Consultant - Asset-Liability Management & Risk | Toronto, Feb 2019 - Sep 2021

- Built and interpreted stochastic economic scenario generators using Monte Carlo simulation to assess interest rate, inflation, FX, and equity risk under base-case and stressed environments for pension and institutional clients.
- Performed asset-only and asset-liability (surplus) portfolio optimization based on VaR and CVaR using Ortec's GLASS platform; conducted risk decomposition, contribution-to-risk attribution, and near-optimal frontier analysis to test allocation robustness.
- Modelled fixed-income and derivatives positions alongside actuarial liabilities; performed cash-flow, liquidity, and hedging analysis across interest rate, inflation, and currency risk for international institutional clients.
- Presented quantitative findings – including duration, currency, inflation, and leverage-overlay strategies – to investment committees at Canadian pension-fund clients.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016